

50.057 (Re-) Introduction to Logic

ISTD, SUTD

Learning Outcomes

- ▶ List the basic elements of Logic
- ▶ Compare the deductive reasoning with the inductive reasoning
- ▶ Explain the need of formal logic
- ▶ List the different types of logic

What is Logic?

Logic is a systematic study of the rules and principles used to distinguish correct reasoning from incorrect reasoning.

- ▶ It is a science of drawing a conclusion given a set of assumptions.
- ▶ It allows to validate whether an argument is valid.

Logic is an Ancient Subject

- ▶ **Trivium** from the ancient Greece (5th - 4th century BC)
- ▶ **Mohist** from the warring states of ancient China (4th - 3rd century BC)
- ▶ **Hetuvidyā** from the ancient India (4th century CE)

Logic forms an important mathematical core of the modern Computer Science.

Why Logic is Useful?

Logic is widely used, in areas of

- ▶ Math proof
- ▶ Medical diagnosis
- ▶ Debugging computer programs
 - ▶ Catching AI hallucination
- ▶ Legal reasonings
- ▶ Playing games
- ▶ Solving puzzles

Why Logic Matters in Agentic Software Engineering?

- ▶ to locate ground truth
- ▶ to develop philosophical views on
 - ▶ what we have been doing w/o LLM
 - ▶ what can we do with LLM
 - ▶ what remains unchanged

Elements of Logic

Logic systems are built up from the following elements, from smallest to the largest

1. Logical sentences (Logical statements)
2. Logical entailment
3. Logical proofs

Logical Statement

A logical statement is the smallest unit in a logical system in the general sense.

A logical statement can be used to describe a possible set of worlds (or instances of the domain).

An Example: Social Media Network

Consider a small social-media network with four friends: Ada, Beatrice, Cara, and Daphne. We are interested in who follows whom.

- ▶ A checkmark in a row records that the friend named at the start of that row **follows** the friend named at the top of the column.
- ▶ The absence of a checkmark means that the corresponding following relationship does not hold.
- ▶ Nobody follows themselves.

One possible world might have

(S1) All other 3 girls following Cara, Cara following them back. Cara does not follow herself.

World A

	Ada	Beatrice	Cara	Daphne
Ada			✓	
Beatrice			✓	
Cara	✓	✓		✓
Daphne			✓	

Let's assume the platform has such an option but Cara decided not to.

Another Possible World

Another possible world might have

(S2) Every friend following exactly one other friend and herself.

World B

	Ada	Beatrice	Cara	Daphne
Ada	✓	✓		
Beatrice		✓	✓	
Cara			✓	✓
Daphne	✓			✓

Note that these are different possible states of the same social-media network given the same statement (S2).

World C

	Ada	Beatrice	Cara	Daphne
Ada	✓		✓	
Beatrice		✓		✓
Cara	✓		✓	
Daphne		✓		✓

And there are many more.

Logical Entailment

Logical entailment is a way to describe how two logical sentences are related to each other.

For example, consider another statement

(S3) Everyone follows at least one person.

We would expect there are many worlds that satisfy (S3).

S2 Entails S3

An observation here is that **every world that satisfies (S2) also satisfies (S3)**.

We say *(S2) logically entails (S3)*.

Questions

1. Does (S1) entail (S3)?
2. Does (S3) entail (S2)?

Logical Proof

A logical proof is a process

1. to draw a conclusion (a logical statement)
2. from a set of premises (a group of logical statements that are connected via conjunction and disjunction)
3. via a sequence of inference steps.

Logical Inference

There are at least 2 kinds of logical inference operations.

1. Deduction
2. Induction

Deductive Reasoning

In deductive reasoning, we move from general statements to specific statements, e.g.

(S4) Adding one to every even number produces an odd number.

We can argue

(S5) $1 + 2$ is an odd number.

Deductive Reasoning (Cont'd)

Another example

(S6) All Term 7 CSD students must have taken Elements of Construction.

and

(S7) All students who take Elements of Construction must have taken Information System and Programming.

Therefore

(S8) All Term 7 CSD students must have taken Information System and Programming.

Inductive Reasoning

In inductive reasoning, we infer general statements from specific statements.

For instance

(S9) I've seen 100 black mynas in Bedok.

Therefore I think

(S10) All mynas are black.

Inductive Reasoning (Cont'd)

This seems to be a valid argument, especially for science and discovery. However it is not very suitable for verification.

Because (S10) might not be valid because I might have missed a leucistic myna.

Hence for logical proof, we largely avoid induction, except for a special instance, namely **structural induction**, which will be discussed shortly.

Fun fact

Modern machine learning is considered a kind of inductive reasoning. Can you explain why?

Logical Proof vs Model Checking

An alternative to logical proof is to enumerate all instances that satisfy the premises and check whether all of them satisfy the conclusion. This process is called **model checking**.

- ▶ One advantage of model checking over logical proof is that it is automatable.
- ▶ One obvious disadvantage of model checking is that it is limited to finite set of instances.

In this module, we put our focus on logical proof.

Why Natural Languages Fail

Natural languages are *ambiguous* by nature. A grammatically correct sentence is ambiguous if there are at least two different interpretations (meanings) of the same syntactic form.

- ▶ There's a girl in the room with a telescope.
 - ▶ a girl is in the room in which the telescope is installed; or a girl holding a telescope in a room?
- ▶ Crowds Rushing to See Pope Trample 6 to Death.
 - ▶ crowds trample 6 or the Pope trample 6?

Why Natural Languages Fail (Cont'd)

- ▶ Scientists Grow Frog Eyes and Ears
 - ▶ Frog eyes and ears are on the scientists' faces; or they are created in the lab by the scientists?
- ▶ Wife tells husband, "go to the supermarket to buy a water melon; if you see eggs, buy ten."
 - ▶ Should the husband come back from the supermarket with one water melon or ten water melons?

To avoid ambiguity, mathematicians and computer scientists use mathematical symbols and operators to form logical sentences. We call it **symbolic logic** or **formal logic**.

Types of Logic

There are many sub-categories of logic, e.g.

1. Propositional Logic
2. Predicate Logic
3. First-order Logic
4. Higher-order Logic
5. ...

We will discuss the first 3 in detail.